

02_Impact_Analysis_CR-014

Change Request: CR-014 – Admin Dashboard Statistics

Project: Student Management System

Impact Analysis Status: Planned / Pre-Implementation

Date: 07 October 2026

1. Change Request Identification

Field	Details
Change Request ID	CR-014
Change Request Title	Admin Dashboard Statistics
Priority	Medium
Difficulty	Medium
Change Type	Enhancement
Implementation Status	Not Started / Planned
Affected Version	Student Management System v1.0 baseline
Primary Area	Administrator Dashboard

2. Change Analysis & Impact

2.1 Change Scope

In Scope

- Add a statistics section to the Admin Dashboard.
- Display Total Students based on current database records.
- Display the number of students registered during the current calendar month.
- Display the number of students registered within the last seven days.
- Display the latest registered student's registration number and name.
- Add a backend dashboard statistics API to provide calculated values.
- Protect the new API using the existing authentication and Admin authorization mechanism.
- Add loading, empty-data, and API-error handling.
- Make the statistics section responsive for desktop, tablet, and mobile screens.

Out of Scope

- Changes to the existing student table structure or student records.
- Changes to student creation, editing, deletion, search, or filtering business rules.
- Changes to the existing login or user-role model.
- Pagination or sorting enhancements to the existing student list.
- Historical analytics, graphs, or reporting beyond the four requested statistics.
- Integration with external analytics or reporting services.

2.2 Assumptions and Constraints

Assumptions

- The existing students table contains the student records required for the statistics.
- The created_at field is available and contains usable registration timestamps.
- The existing Admin authentication and authorization middleware is working correctly.
- The existing frontend can call the Node.js/Express backend through the configured API client.

- The current student dataset is small enough that aggregate dashboard queries will have low performance impact.

Constraints

- The implementation should reuse the existing database structure where possible.
- The statistics endpoint must not expose administrative information to unauthorized users.
- Existing student management functionality must continue to operate without regression.
- The dashboard must remain usable when statistics are unavailable or the student dataset is empty.
- The change is limited to the approved CR-014 scope and should not introduce unrelated functionality.

2.3 Impact Analysis

Identify the areas that could be affected by the change.

Area	Impact	Details
Frontend	Medium	DashboardPage.jsx will be enhanced to request statistics and display four summary cards, loading state, and empty/error states.
Backend	Medium	A new dashboard statistics controller and route will calculate and return student statistics.
Database	Low	Existing student records will be queried using COUNT, date-based filtering, and latest-record retrieval. No schema change is expected.
API	Medium	A new authenticated GET dashboard statistics endpoint will be introduced without changing existing student API contracts.
Authentication / Authorization	Low	The new endpoint will reuse the existing authentication and Admin-role authorization controls.
Security	Medium	Server-side access control is required so that unauthenticated and non-admin users cannot access dashboard statistics.
User Interface / Styling	Medium	New responsive statistics cards and related styling will be added to the Admin Dashboard.
Configuration	Low	No major configuration change is expected; the existing backend/frontend API configuration will be reused.
External Services	None	No external service or third-party integration is required.
Performance	Low	Additional aggregate queries will run when the Admin Dashboard loads. The expected impact is low for the current system size.
Testing	Medium	Functional, API, authentication, authorization, negative, error-handling, responsive UI, and regression testing are required.
Documentation	Medium	The CR-014 implementation and testing evidence must be documented in the related change report.
Other	Low	No other significant system area is expected to be affected.

2.4 Affected Components and Files

Component / File	Type	Changed?	Purpose / Reason for Change
frontend/src/pages/DashboardPage.jsx	Frontend	Yes	Request dashboard statistics and display the four statistics cards, loading state, and error/empty handling.
frontend/src/styles/dashboard.css	Frontend / UI	Yes	Add responsive styling for the statistics section and cards.
backend/src/app.js	Backend / API	Yes	Register the new dashboard statistics route.
backend/src/controllers/dashboardController.js	Backend	Yes	Calculate total, monthly, seven-day, and latest-student statistics.
backend/src/routes/dashboardRoutes.js	API	Yes	Expose the protected GET /api/dashboard/stats endpoint.
Existing authentication middleware	Authentication / Authorization	No / Reused	Existing middleware will be reused to protect the new Admin-only endpoint; no new authentication model is required.
Existing students table	Database	No schema change	Existing student records and created_at data will be queried; no migration is expected.

3. Overall Impact Summary

CR-014 is expected to have a Low-to-Medium overall impact on the Student Management System. The main affected areas are the Admin Dashboard frontend, the backend dashboard API, API access control, and database read queries. No database schema migration is expected, and existing student CRUD, search/filter, authentication, and authorization functionality should remain unchanged.

The principal implementation risks are incorrect statistics, unauthorized API access, dashboard errors when data is unavailable, and minor performance overhead from additional database queries. These risks will be controlled through server-side calculations, Admin-only authorization, error handling, responsive UI validation, and regression testing.

4. Impact Decision

Decision Area	Assessment
Overall Impact	Low to Medium
Database Migration Required	No – expected
Existing API Contract Breaking Change	No
Authentication Model Change	No
Authorization Change	New endpoint protected by existing Admin authorization
External Service Dependency	No
Regression Testing Required	Yes
Implementation Approval	Suitable for implementation within CR-014 scope